ELSHADAI TEGEGN

Software Engineer

+19735835012 | Email | LinkedIn | Github | Website

WORK EXPERIENCE

SOFTWARE ENGINEER AT AUDIBLE/AMAZON — *NJ, US* *JULY 2024 - Present*

- Led the implementation of a new feature on Product Detail Page (PDP) that increased user sign-ups by 2% within two weeks, generating an additional $415K in revenue.
- Proactively increased branch coverage for four core packages to 95% during the gap week between Q3 and Q4, significantly improving code reliability and maintainability.
- Deliver high-impact features across the stack, collaborating with cross-functional teams to enhance user experience and meet business objectives.

TEACHING ASSISTANT AT IOWA STATE UNIVERSITY — *Ames, IA* *JAN 2024 - May 2024*

- Assisted in teaching COMS 227 (Object-Oriented Programming), leading lab sessions and office hours to support student learning.
- Provided guidance on programming concepts, debugging, and best practices in Java.
- Graded assignments and exams, offering constructive feedback to enhance student understanding.
- Collaborated with faculty to refine course materials and improve instructional effectiveness.

SOFTWARE ENGINEER AT ORGANIZATION OF SOUTHERN COOPERATION — *ETHIOPIA* *OCT 2023 - DEC 2023*

- Enhanced digital access to education and research by working on the Greater South Information System (GreSIS).
- Engineered user access controls for GreSIS, bolstering security and user experience.
- Implemented social networking features for academic collaboration on the platform.

MACHINE LEARNING RESEARCH INTERN AT INSTADEEP — *UK* *JAN 2023 - JULY 2023*

- Engineered the implementation of a Rainbow DQN algorithm in Mava, optimizing model training efficiency by 30% and enhancing multi-agent environment capabilities.
- Crafted robust reinforcement learning environments in Jumanji, including a Graph Coloring environment and a Sliding Tiles Puzzle environment, which resulted in a 20% improvement in simulation fidelity and reliability across multiple testing scenarios.

SENIOR BACKEND ENGINEER AT GEBEYA INC. — *ETHIOPIA* *MARCH 2023 - JULY 2023*

- Spearheaded the architecture and development of critical backend functionalities, comprising 48% of the system, for Mesirat, a platform designed to connect 1,000 gig market service providers with top talent.
- Assembled a user authentication system by collaborating with diverse teams, bolstering platform security, and reducing unauthorized access by 85%.
- Streamlined data access by implementing an optimized caching solution, cutting data retrieval times by 40%.

SOFTWARE ENGINEERING INTERN AT AMAZON, AWS — *GERMANY* *OCT 2022 - DEC 2022*

- Worked on an end-to-end service to annotate images and browse image labels
- Implemented a frontend SPA using AWS' internal UI framework and a backend using AWS Lambda.
- Created the AWS CDK typescript code to deploy both the front-end and backend and provision additional AWS resources like connections to OpenSearch

SOFTWARE ENGINEERING INTERN AT AMAZON, AUDIBLE — *NJ, US* *JUNE 2022 - AUG 2022*

- Spearheaded the development of payment-related features, prioritizing secure transaction environments and compliance adherence.
- Amplified user sign-ups by 16K annually through strategic full-stack feature development, blending user-friendly design with functional excellence.

AUTOMATION TESTER AND DIGITIZATION ENGINEER. — *POLAND* *AUG 2021 - MARCH 2022*

- Executed automation testing using Selenium and Python for the Plume Mobile application, ensuring robust and reliable application performance.
- Championed the digitization of company processes, augmenting process flow speed by 21% through strategic technological interventions.

COO, CO-FOUNDER, FULL-STACK ENGINEER AT KNOVUSLAB. — *ETHIOPIA* *MARCH 2020 - DECEMBER 2023*

- Pioneered the development of Motorride, a ride-hailing platform mobile app, overseeing all phases from concept to deployment and contributing to mobility solutions.
- Fullstack development of Shega, Ethiopia's prominent tech news platform
- Orchestrated the system architecture, development, and leadership of the backend team for NileStream, an indigenous audio streaming platform, navigating through the challenges of the tech start-up ecosystem.

PUBLICATIONS

Bonnet, C., Luo, D., Byrne, D.,... Tegegn, E., et al. "**Jumanji: a Diverse Suite of Scalable Reinforcement Learning Environments in JAX.**" ArXiv, ICLR 2024.

PROJECT EXPERIENCE

RESEARCH INTERN AND MENTOR AT MIT SGI PROGRAM **Remotely**

July - Aug | Intern - 2022 | Mentor - 2023 & 2024

Merged roles of mentor and researcher, contributing to projects in computational geometry and enhancing methodologies in computer graphics.

- How to Mend a Broken Part: Led as a mentor to innovate in geometric fracture assembly and formulate an evaluation metric, inspiring an open-source assembly challenge.
- Plane and Edge Detection (RANSAC): Contributed to meticulous 3D point cloud plane detection, leveraging the RANSAC algorithm for structured and precise data processing.
- Revisiting Computational Caustics: Worked with SGI Fellows, exploring differential rendering and optimal transport solutions, streamlining reflection calculations via vectorization, and addressing mesh reconstruction amidst noise.
- Out-of-Context Data Augmentation for 3D Scenes

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY **GA, US, 2025-Ongoing**

- *MSC in CS, Specializing in ML*

IOWA STATE UNIVERSITY **IA, US, 2024-Paused**

- *PhD in CS*

ADDIS ABABA UNIVERSITY **Addis Ababa, Ethiopia, 2018-2023**

- *Bachelor of Science, Software Engineering*

  M.GPA: 3.67

MAX PLANCK INSTITUTE **SAARBRÜCKEN GERMANY, 2022**

- *CMMRS Pre-doctoral School*

TENNESSEE TECHNOLOGICAL UNIVERSITY **TN, US, 2022**

- *Global UGRAD scholarship exchange program*

PROFESSIONAL DEVELOPMENT, LEADERSHIP AND CERTIFICATIONS

SHARE PROGRAM - CERTIFICATE **ETHIOPIA, March 2020 - Jan 2022**

Served as President of ShARE AAU/AAiT 2022 Program, consulting on strategic initiatives for Axiom Financials and conducting mobility research post-COVID-19 for Valeo.

SPECTRUM MANAGEMENT - CERTIFICATE **KENYA, AUG 2023**

AWS ML/AI NANODEGREE SCHOLARSHIP PROGRAM - CERTIFICATE **ETHIOPIA, NOV 2023**

SKILLS

Languages: Dart (Flutter), C++, JavaScript (NodeJS, ReactJS, ExpressJS), C# (.NetCore), Python (Django, Flask, Matplotlib, Seaborn, Pandas), Java, Kotlin
ML & DL: TensorFlow, Jax, Pytorch (Neural Networks, NLP, Deep Learning, CNN)
Front-end: HTML, CSS, Bootstrap
Database Management: Oracle, PostgreSQL, MySQL, MongoDB
Other: AWS Cloud, Docker

AWARDS

- Huawei ICT Innovation 2021, 2nd place from Ethiopia: CERTIFICATION
- Hult Prize Regional, 2nd place winner
- Ethiopian Collegiate Programming Competition, top 15% from East Africa and Arab: CERTIFICATION